

Guanhua Zhu

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Google Scholar | GitHub: MrFantasyZ | New Jersey Institute of Technology, Newark, NJ

Research Interests

My research builds **structured knowledge representations** — knowledge graphs, hypergraphs, and graph neural networks — and uses them to make large language models and multimodal learning systems more reliable, particularly in healthcare settings. Three threads run through my work:

- **Structured Retrieval-Augmented Generation.** Aligning how a corpus is indexed with how a question is decomposed. In HyperPathsRAG, n -ary hyperedges replace binary graph edges so that facts survive indexing intact, and retrieval returns ordered reasoning paths rather than an unordered top- K context — improving multi-hop QA across three benchmarks and three backbone LLMs.
- **Self-Evolving LLM Agents and Agent Memory.** Giving agents a memory with a typed abstraction hierarchy that maintains itself. STRATA organizes experience into a three-layer graph (skills / workflows / meta-frameworks) joined by typed edges, with reflection, community-induced abstraction, pressure-triggered garbage collection, and contrast edges that suppress negative transfer — attaining the best average score at the lowest latency and context cost among memory-augmented agents.
- **Graph and Multimodal Learning for Biomedical Data.** Carrying the same structural view into health informatics: graph-enhanced multimodal fusion for human motion tracking, and region-aware evaluation plus latent-diffusion harmonization of multi-site 3D brain MRI, so that downstream biomarkers reflect biology rather than scanner artifacts.

Earlier work applied graph neural networks, sparse wavelet decomposition, and reinforcement learning to signal-level fault diagnosis and control, which is where my interest in graph-structured representations began.

Education

New Jersey Institute of Technology

Ph.D. in Computer Science; Advisor: Prof. Chenxi Yuan

Newark, NJ

Jan. 2025 – Present

Purdue University Northwest

M.Phil. in Electronic and Computer Engineering, GPA: 3.9/4.0

Hammond, IN

Jan. 2021 – Mar. 2023

Tianjin University of Technology

B.S. in Mechatronics Engineering, GPA: 3.2/4.0

Tianjin, China

Aug. 2016 – Jun. 2020

Publications

Peer-Reviewed

- [1] **G. Zhu**, Z. He, M. Du, C. Yuan. “HyperPathsRAG: Multi-Target Path Approximation over Hypergraph Knowledge Graphs for Multi-Hop QA.” *Findings of the Association for Computational Linguistics: EMNLP*, 2026.
- [2] D. Yan, **G. Zhu**, Y. Chen, J. Tao, Y. Ding, X. Zhang, C. Yuan, W. Yin. “Advancing Human Motion Tracking via a Graph-Enhanced Multimodal Learning Model.” *IEEE Engineering in Medicine and Biology Conference (EMBC)*, 2026. (Poster)
- [3] **G. Zhu**, X. Zhou, L. Tan. “Deep Learning Based Bearing Fault Detection Using Frequency Resampling for Different Shaft Rotating Speed.” *IEEE Access*, 2024.
- [4] X. Zhou, **G. Zhu**, L. Tan. “Reinforcement Learning Assisted Design for FOPID Control.” *IEEE International Conference on Electro Information Technology (EIT)*, 2024.
- [5] **G. Zhu**, X. Liu, L. Tan. “Bearing Fault Diagnosis Based on Sparse Wavelet Decomposition and Sparse Graph Connection Using GraphSAGE.” *IEEE International Conference on AI in Cybersecurity (ICAIC)*, pp. 1–6, 2023.

Under Review

- [6] **G. Zhu**, et al. “STRATA: Self-Maintaining Structured Experience Graphs for Self-Evolving Language Agents.” (*Under review, AAAI*)
- [7] J. Duan*, **G. Zhu***, X. Shi, M. Schnaider Beerli, K. Xu, C. Yuan. “Joint Global and Parcellation Level Harmonization of Multi-Site 3D MRI Data.” (*Under review, IEEE Journal of Biomedical and Health Informatics*)
*Equal contribution.

Research & Professional Experience

Graduate Research Assistant, New Jersey Institute of Technology

Advisor: Prof. Chenxi Yuan

Newark, NJ

Jan. 2025 – Present

- Designed **HyperPathsRAG**, a hypergraph-based RAG framework that decomposes questions into n -ary sub-questions, jointly matches them to retrieve reasoning paths, and scores paths by chain completeness; best average Exact Match on every backbone across HotpotQA, 2Wiki, and MuSiQue (up to 42.2 EM / 56.2 F1 with Llama-3.3-70B-Instruct).
- Built **STRATA**, a self-maintaining three-layer experience graph for LLM agents, evaluated on AppWorld, WebArena, ALFWorld, and ScienceWorld against seven baselines and two frozen backbones; contrast edges removed catastrophic negative transfer, lifting ScienceWorld from 78.7 to 96.7 within the same system.
- Co-led **GrADHarm** and the **Regional Wasserstein Distance (RaD)** metric for parcellation-level harmonization of multi-site 3D MRI (SRPBS, IXI), improving local anatomical fidelity over GAN- and diffusion-based baselines.
- Contributed graph-enhanced multimodal fusion for human motion tracking (IEEE EMBC 2026).

AI Algorithm Engineer, Meituan

Chaoyang District

Mar. 2024 – Dec. 2024

Beijing, China

- Developed an LLM fine-tuning pipeline for a production question-answering chatbot.
- Delivered a service-recommendation push system built on knowledge graphs and graph neural networks.

Research Assistant, Purdue University Northwest

ECE Department; Advisor: Prof. Li-zhe Tan

Hammond, IN

May 2022 – May 2023

- Developed a graph-neural-network method for bearing fault diagnosis (ICAIC 2023).
- Developed a bearing fault identification method combining sparse wavelet decomposition with frequency-domain resampling to generalize across shaft speeds (IEEE Access 2024).
- Developed a reinforcement-learning-based tuning method for FOPID control (EIT 2024).

Research Assistant, Center for Innovation through Visualization and Simulation

CIVS, Purdue University Northwest

Hammond, IN

May 2022 – May 2023

- Constructed a medical knowledge graph in Python and Neo4j.
- Built a medical question-answering system on top of that knowledge graph.

Course Grader, Purdue University Northwest

ECE Department

Hammond, IN

May 2022 – May 2023

- ECE 42301 Power Electronics and ECE 20100 Linear Circuit Analysis (Prof. Dave Kozel); ECE 31100 Electric and Magnetic Fields (Prof. Constantin Apostoiaia).

Skills

Programming	Python, Java, MATLAB, C/C++, R, SQL, HTML
ML / LLM	PyTorch, TensorFlow, Hugging Face, LangChain, OpenAI API, vLLM, PyTorch Geometric
Data / Graph	Neo4j, Py2neo, PySpark, FAISS, BeautifulSoup
Tools	Linux (CentOS, Ubuntu), Git, Docker, L ^A T _E X, VS Code, PyCharm, Colab
Languages	Chinese (native), English (GRE: 328, TOEFL: 97)